

SSTI Summer Internship



UNITED STATES
SPACE FORCE

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The University of Texas at Austin

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Intern: Guangzhi (Simon) Chen





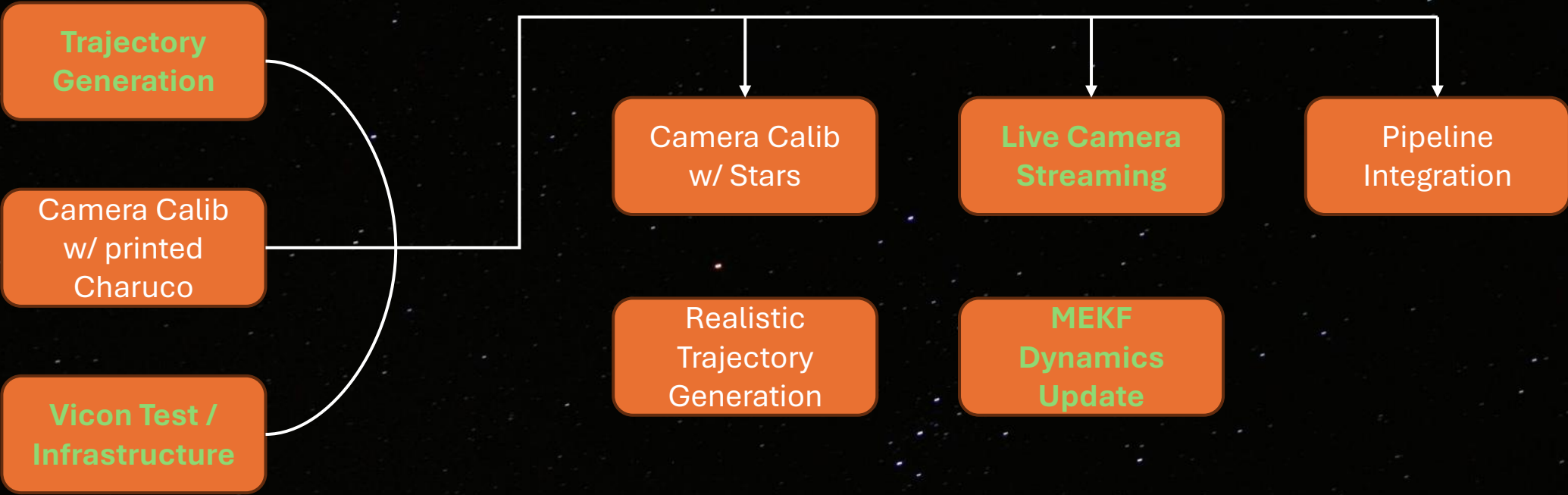
- Guangzhi Chen
- UT Austin '28
- BS Aerospace Engineering
- Motivation: I've mostly done the controls side of GNC, so I wanted to explore guidance and navigation. Figuring out where something is, with error and uncertainty, seemed like a really cool problem.

Key Personnel

• UTEP: Velez-Reyes, Najera, Roberts + students

• UT Austin: Pryor, Akella + students

Improving the data-collection pipeline and the pose-estimation filter for spacecraft rendezvous



Trajectory Generation

Context: Enable us to more easily create trajectory to get imagery for camera calibration, Vicron rough pose estimation, and Megapose refinement pipeline

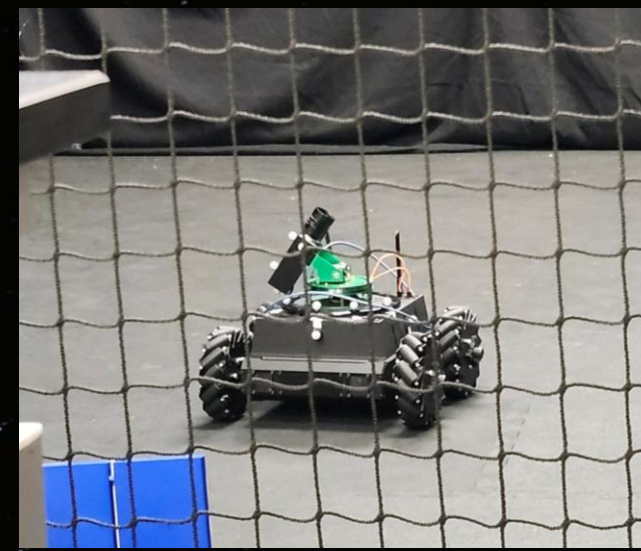
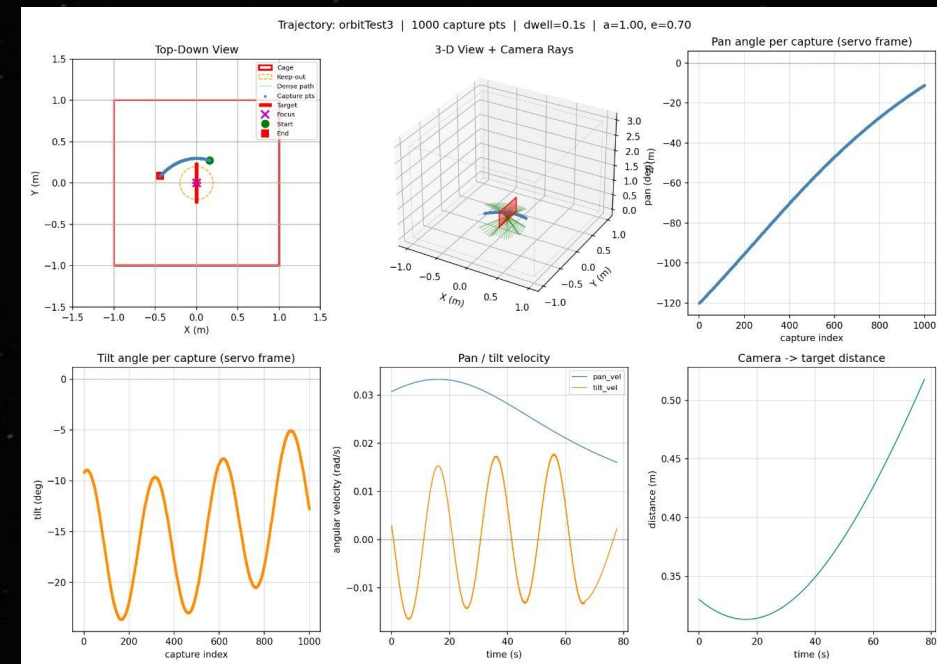
Issues with the script:

- Target-location CLI argument not working
- CSV output unusable → Windows CRLF line endings vs Unix
- Arc could only face +x direction
- Did not need to use “shrinking concentric arcs” feature
- Could only create circular arcs
- Target_buffer was a line boundary instead of a shape

New Script using Classical Orbital Elements:

- $a \rightarrow$ semi-major axis (size), $e \rightarrow$ eccentricity, $\Omega \rightarrow$ RAAN (rotation about z), $\theta \rightarrow$ true-anomaly (select which arc segment)
- Added camera tilt for more view variation for training data
 - Adds a sine wave to the base tilt and fixed a bug that would break the gradient-calculations if dwell time was applied
- Can be extended to 3D
- Can set a direction the chassis is facing

Issue remaining: Found an apparent $\sim \pm 90^\circ$ pan limit; at 90° the camera tilt straight up



Vicon Testing

Issue Encountered:

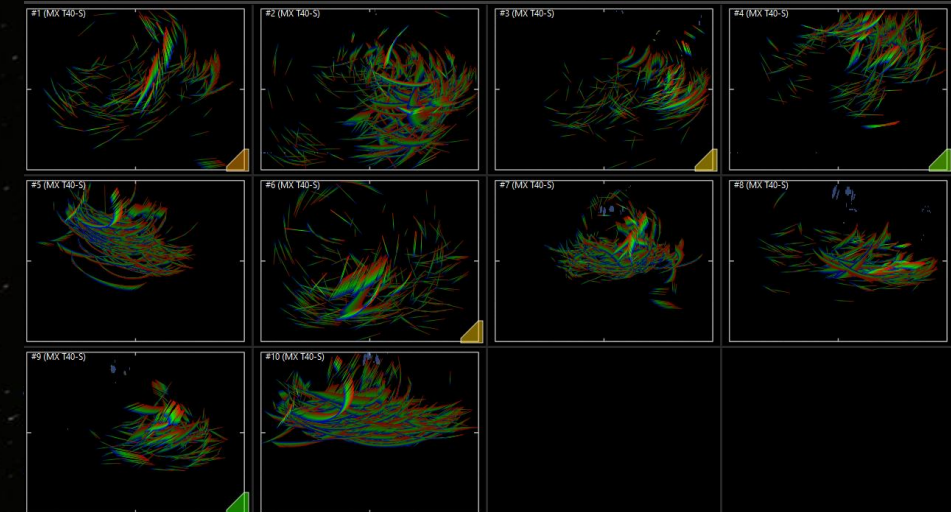
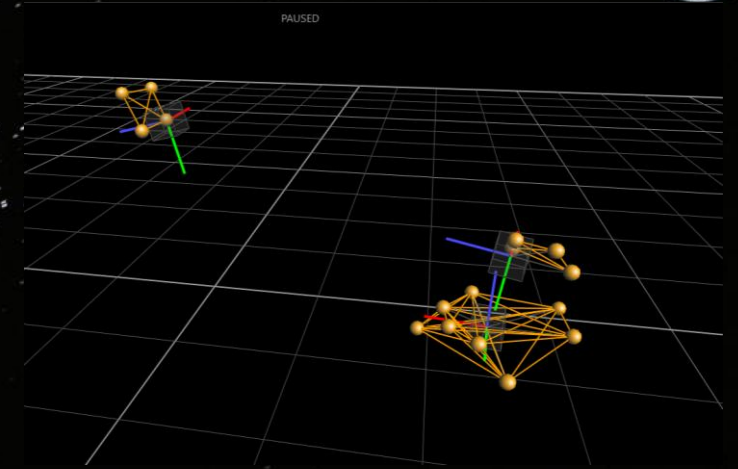
- During data collection, tracked object jittering/teleporting around the cage while stationary

Resolution:

- Thought it was bad calibration so recalibrated the cage
- Added more reflective-markers on Eowyn as it was too similar to the robot carrying Soho

Other Task:

- move to Anna Hiss Gym: rebuilt all objects, learned to redefine the camera object's coordinate frame



Live Camera Streaming

Context: As we try to get data for different trajectory, we need to manual change the focus and aperture. However current methods require us to spin down recording node, broadcasting node, and other nodes if active. This node allows us to receive images while other nodes are running

Constraint: Streaming raw image directly will clog the network. Additionally, some topics use best-effort QoS (floods the network with messages and if not picked up, it will silently drop messages)

Solution:

- Design: A separate node instead of a different driver that communicates with the camera
 - The driver has a built-in image compressor. however, someone added a rule if you turn resizing ON, you must turn the full-size raw output OFF
- New stream_compressor node: configurable target width, JPEG quality, stream rate, QoS

Issues Remaining:

- Raw image should not cross over the network (need more verification)
- Bandwidth of the network or send/receiver is small as image is target width as 1/8 and JPEG Quality is 50%



MEKF Dynamics Upgrade

Context: The current translation part model was a Discrete Wiener-Process Acceleration (DWPA) model. It tracks position/velocity/acceleration per axis and treats acceleration as a gravity-blind random walk. It can't use physics to smooth noise and its uncertainty grows unbounded during measurement gaps.

New Translation State: $\delta\alpha = [\delta a, \delta\lambda, \delta e_x, \delta e_y, \delta i_x, \delta i_y]$

- 6 nonsingular ROE (Koenig/Guffanti/D'Amico 2017; the states Park & D'Amico 2023 use)
- $a \cdot \delta\alpha$ in meters (SHIRT convention and better matrix conditioning)
- Full State: $\mathbf{x} = [\delta\mathbf{g}^T, \omega^T, \alpha^T, \delta\alpha^T]^T$

Why ROE:

- Under Keplerian motion, STM is exact and eccentricity-independent: one nonzero entry, $\dot{\delta\lambda} = -\frac{3}{2}\delta a$
- Can work for circular and elliptical orbits, less constraints than Clohessy–Wiltshire Equations

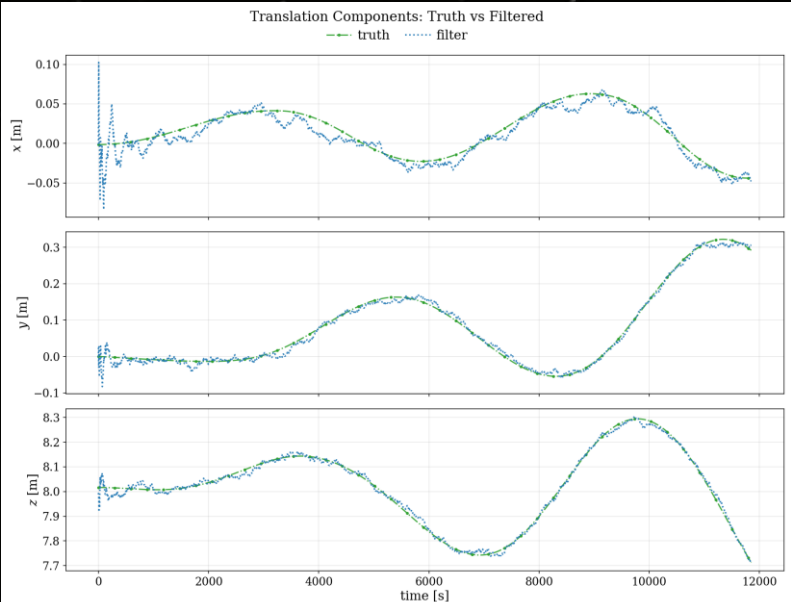
MEKF Dynamics Upgrade (cont.)

SHIRT Validation

Measurement:

- Issue: States in ROE and measurements in camera frame
- Solution: convert states to measurement space
 - $h(x) = R_{body \leftarrow RTN}(t) \cdot J_{roe2rtn} \cdot a\delta\alpha$
- Assuming body = camera frame and point object, we get $R_{body \leftarrow RTN}(t)$ from SHIRT
- $J_{roe2rtn}$: (computed by central finite difference)
 - Get $\delta\alpha$ from $a\delta\alpha / a$
 - Use ROE equations to get target_OE
 - Use Target_OE and Chaser_OE to get to ECI frame and find the difference to get $\delta r_{ECI} (r_d - r_c)$
 - Then use $R_{RTN \leftarrow ECI}$ to get to δr_{RTN}
 - Perturb each $a\delta\alpha$ state with ± 1 meter
- Issue: Initialization of filter but measurement only give XYZ in camera frame
- Solution: Set $a\delta\alpha = 0$ and P to have big covariance on the diagonal. Then run one update with Kalman gain function and covariance of P in the observable axis shrink

Roe1	DWPA	ROE
Measurement error	0.127 m	0.127 m
Filter error (mean)	0.122 m	0.017 m
Filter error (max)	0.334 m	0.118 m
RMSE	0.132 m	0.020 m



MEKF Dynamics Upgrade (cont.)

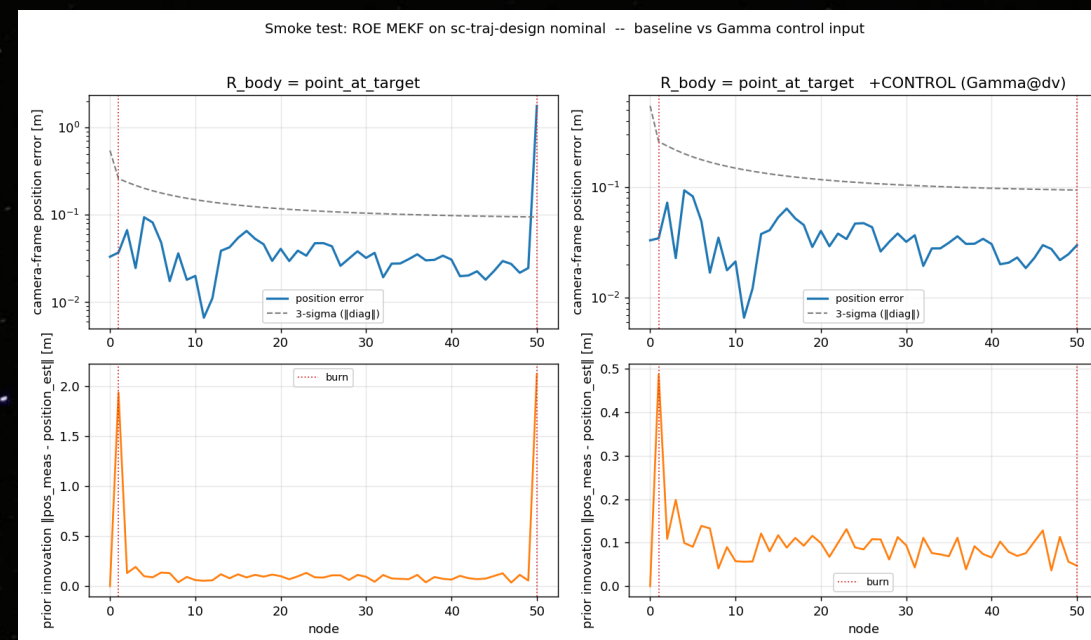
Issue: Trajectories for Rendezvous and Proximity Operation tend to have delta v's to approach the target, so we need to model that in our filter or it will appear as a large "innovation"

Solution: Find a control input matrix that tells us how the ROE changes given a delta v

- $\Gamma_{\text{analytical}} = a_c \cdot \frac{\partial \delta \alpha}{\partial \text{OE}} \cdot \frac{\partial \text{OE}}{\partial v_{\text{RTN}}}$
 - However, I could not find an analytical solution for our nonsingular ROE, most are for quasi non-singular ROE
 - Compute numerically, by finding the RTN vectors in ECI frame and adding dv in the directions of the RTN vectors to see how the orbital elements change
- Currently, only have instantaneous thrust modeled

Validation: Using trajectory created by sc-traj-design repo

- End-to-end filter run, baseline vs control: at the final burn, error 1.77 m $\rightarrow$ 0.03 m, innovation 2.12 m $\rightarrow$ 0.05 m;



Potential Future Work

- Extend trajectory generation to 3D for the SENSS Lab
- Add J_2 /drag perturbations to the filter dynamics
- Upgrade the attitude dynamics with a physics-based rotational model
- Continuous thrust modeling

Thank You!

Thank you to Anand Agrawal and Dr. Akella for the mentorship and opportunity